Smog Watch Pakistan

Project Proposal (P02)

DS-311 Data Mining | BS-AI 2024F, Section A | NBC, NUST | Fall 2026

Instructor: Dr. Zirak Khan | Lab: Syed Daniyal Asghar

Project Overview

Smog is a recurring winter health hazard in Pakistan, yet most residents see only a single daily AQI number with no context about *why* it spikes or whether tomorrow will be worse. This project builds an end-to-end data-mining pipeline on open data: hourly $PM_{2.5}$, PM_{10}, NO_2, O_3 and US AQI from the Open-Meteo Air Quality API (CAMS global model, Aug 2022 onward), integrated on timestamp and city with temperature, wind, humidity and precipitation from the Open-Meteo Historical Weather API. We cover six cities (Lahore, Karachi, Islamabad, Peshawar, Faisalabad, Quetta), aggregate to daily records, and mine them with clustering, outlier detection and classification. Results are served in a Streamlit dashboard with a city selector, AQI calendar heat-map, pollution regime, next-day risk band and anomaly timeline.

Team and Roles

Member	Reg. No.	Lead Role	Responsibilities and mining technique
Inshan Haider	000000	Data Lead	API acquisition, integration, data-quality report, EDA; outlier/episode detection
Muhammad Obaidullah	000000	Mining Lead	Feature engineering, next-day AQI classification (DT/RF), evaluation and leakage checks
Shayan Afzal	000000	Product Lead	Pollution-regime clustering (K-means, hierarchical), Streamlit app, deployment, report

Problem Statement

Modelled pollution data are freely available but rarely turned into actionable insight for Pakistani cities. We want to identify which weather and seasonal conditions are associated with smog episodes, group days into interpretable pollution regimes, and test whether abnormal pollution days can be flagged one day in advance using only information available today. The work must be reproducible on free Google Colab, use time-based validation, and state clearly that CAMS values are modelled rather than measured.

Research Questions

RQ1. Drivers. How do wind speed, temperature, humidity, precipitation, season and hour of day relate to $PM_{2.5}$ levels in each city, and is winter $PM_{2.5}$ significantly higher than in the rest of the year (hypothesis test)?

RQ2. Regimes. Can daily pollutant and weather profiles be grouped into distinct pollution regimes (K-means versus hierarchical clustering), and how do these regimes differ across cities and months?

RQ3. Early warning. Can next-day AQI category be predicted with a Decision Tree and Random Forest under a time-based split (macro-F1 versus a persistence baseline), and which days do z-score/IQR, DBSCAN and Isolation Forest flag as abnormal?

Plan

Stages follow the course pipeline: acquire and integrate (Colab), data-quality report, EDA, transformation, mining with baselines, evaluation, interpretation, and web app. Code and notebooks are kept in a shared GitHub repository to which all three members commit. Optional expert module: 24–72 h $PM_{2.5}$ forecasting or a CAMS versus OpenAQ ground-sensor bias study.